

Training-Stage Connectivity in VGG11 on CIFAR-10

Experimental results and interpretation

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Abstract

This report asks how compatibility between independently trained networks changes during training. Three pairs of VGG11 models are compared at twelve checkpoints using symmetry-aligned linear interpolation and optimized nonlinear paths. Linear incompatibility appears within the first few epochs, permutation alone leaves large barriers, and positive scaling removes most of the remaining worse-endpoint barrier. Optimized Bézier and polygonal paths remove more than 99% of the raw test-loss chord barrier on average. The conclusions are empirical statements about the tested alignment algorithms and path families, rather than claims about all possible transformations or paths.

1 Question and experimental idea

Standard mode-connectivity studies compare converged solutions. Here we ask when two independent runs become incompatible. Since an early checkpoint is not necessarily a local minimum, we call this *training-stage connectivity*. For checkpoints $\theta_A^{(n)}$ and $\theta_B^{(m)}$, we study:

- same-stage pairs A_n-B_n ;
- cross-stage pairs $A_{200}-B_n$ and A_n-B_{200} ;
- within-run controls A_n-A_{200} and B_n-B_{200} .

Six independently trained no-BN VGG11 models are grouped into three seed pairs. We use epochs $\{0, 1, 2, 3, 5, 10, 20, 30, 60, 100, 150, 200\}$, treating epoch zero as an initialization control.

The linear experiment compares raw interpolation, weight matching, Sinkhorn, and positive-scaling extensions of both alignment families. Validation data select the best permutation-only method and the best method overall for each ordered checkpoint pair. These choices are then frozen and evaluated on two additional seed pairs.

The nonlinear experiment keeps the raw endpoints fixed. A quadratic Bézier path is fitted for every same-stage, cross-stage, and within-run pair. Difficult pairs receive additional Bézier and two-bend polygonal restarts. Validation data select the final path before test evaluation.

2 Two complementary barrier definitions

For a path $\gamma(\lambda)$ with endpoint costs $C(\theta_A)$ and $C(\theta_B)$, the conventional chord barrier is

$$B_{\text{chord}} = \sup_{\lambda \in [0,1]} [C(\gamma(\lambda)) - ((1-\lambda)C(\theta_A) + \lambda C(\theta_B))]. \quad (1)$$

It measures the largest deviation above the line joining the endpoint costs. We also report

$$B_{\text{worse}} = \sup_{\lambda \in [0,1]} C(\gamma(\lambda)) - \max\{C(\theta_A), C(\theta_B)\}, \quad (2)$$

which asks whether the path becomes worse than the worse endpoint.

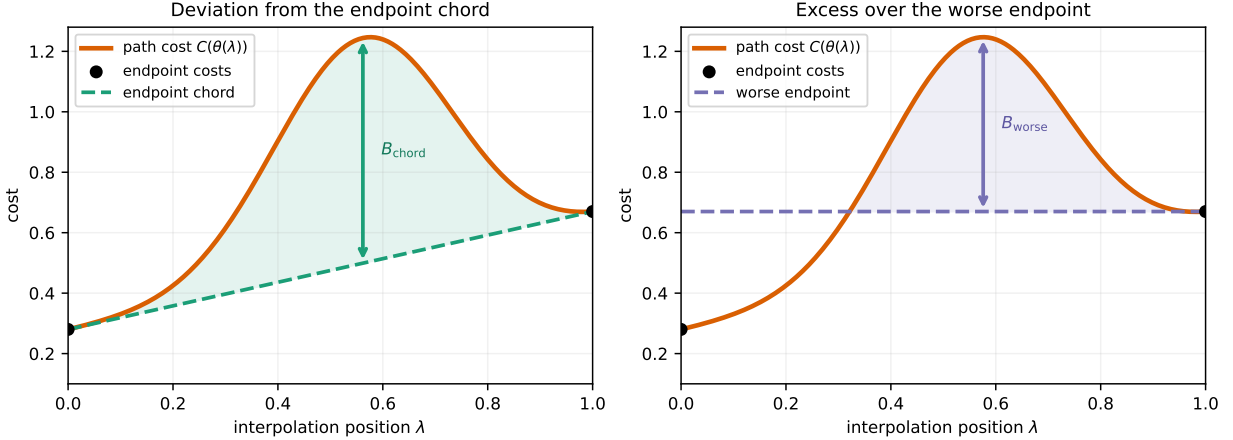


Figure 1: The same path evaluated by B_{chord} and B_{worse} . The distinction matters most when the endpoint losses are unequal.

Both quantities are needed. For example, a random early endpoint can make B_{worse} zero simply because it already has high loss, while B_{chord} still detects curvature above the endpoint trend. Conversely, B_{chord} does not directly answer whether the path is worse than either endpoint. Both metrics are sampled estimates and should be interpreted together with absolute loss.

3 Linear stage connectivity

3.1 Same-stage results

Figure 2 compares the best permutation-only method with the best overall method, which may use positive scaling. A large permutation-only barrier appears almost immediately: mean test-loss B_{worse} is 0.375 after one epoch and peaks at 0.834 at epoch 5. It then declines to 0.251 at epoch 200. Linear incompatibility therefore develops early and is not monotonic throughout training.

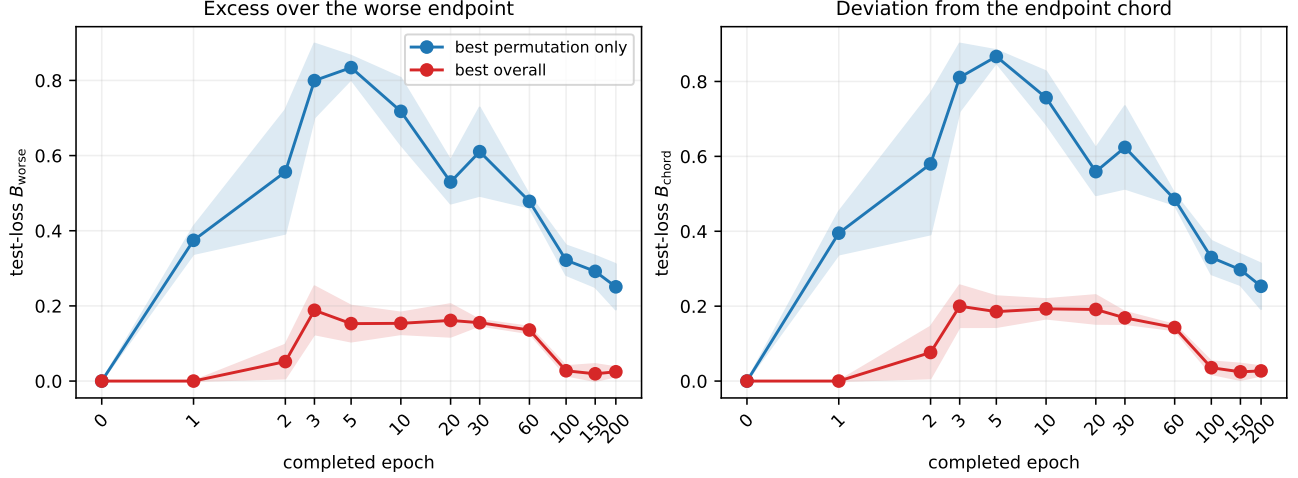


Figure 2: Same-stage linear connectivity. Both panels report test loss; the left uses B_{worse} and the right uses B_{chord} . Curves show mean $\pm$ one standard deviation across three seed pairs.

Positive scaling produces the main improvement. The best overall B_{worse} is zero through epoch 1, rises to 0.188 at epoch 3, stays near 0.14–0.16 through epoch 60, and falls to 0.025 at epoch 200. Every validation-selected overall method is scale-aware: Sinkhorn plus scale refinement is selected for 91 of 144 ordered cells, WM plus scale for 52, and joint Sinkhorn plus scale once. Among permutation-only methods, Sinkhorn wins 86 cells and WM 58.

Table 1: Selected same-stage test-loss barriers, mean $\pm$ standard deviation over three seed pairs.

Epoch	Permutation-only B_{worse}	Overall B_{worse}	Overall B_{chord}
0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
1	0.375 $\pm$ 0.037	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
3	0.800 $\pm$ 0.099	0.188 $\pm$ 0.064	0.200 $\pm$ 0.056
5	0.834 $\pm$ 0.032	0.153 $\pm$ 0.048	0.185 $\pm$ 0.041
60	0.478 $\pm$ 0.017	0.136 $\pm$ 0.009	0.143 $\pm$ 0.007
100	0.322 $\pm$ 0.039	0.027 $\pm$ 0.012	0.036 $\pm$ 0.017
200	0.251 $\pm$ 0.061	0.025 $\pm$ 0.012	0.027 $\pm$ 0.012

Across the complete checkpoint matrix, the mean test-loss B_{worse} falls from 0.431 with the best permutation-only alignment to 0.0408 with the best overall method, a 90.5% reduction. Mean test-error B_{worse} similarly falls from 15.31 to 1.48 percentage points. Positive scaling therefore explains a large part of the mismatch left by permutation.

3.2 Cross-stage results

Figure 3 shows final/intermediate pairs. For $n \in \{0, 1\}$, B_{worse} is zero in both directions, but B_{chord} remains between 0.30 and 0.70. The early endpoint provides such a high baseline that the interpolation does not exceed it, even though the path rises well above the endpoint chord. By epochs 60–200, the two definitions and both directions are much closer.

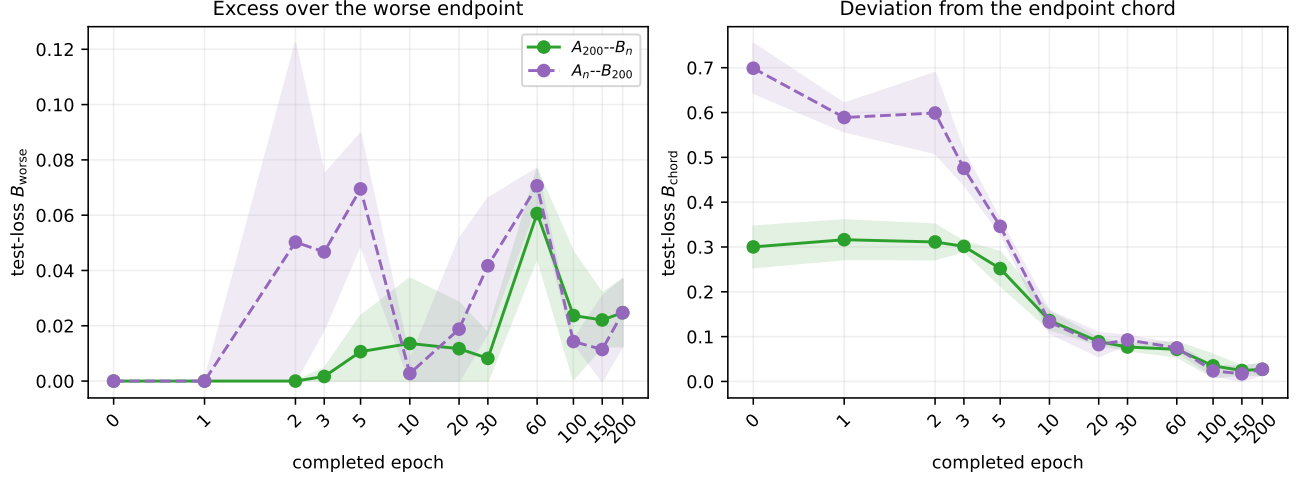


Figure 3: Cross-stage linear connectivity after the best overall alignment. Both panels report test loss and show mean $\pm$ one standard deviation over three seed pairs.

The cross-stage result is the clearest reason not to report only one barrier definition. A statement that early/final pairs have “zero barrier” is true under B_{worse} but incomplete under B_{chord} .

4 Nonlinear stage connectivity

The nonlinear experiment contains 168 primary paths across the three seed pairs. Figure 4 compares raw linear interpolation with the validation-selected curved path. For independent same-stage models, the mean raw test-loss B_{chord} is 1.495, while the selected nonlinear mean is 0.0029, a 99.8% reduction. The corresponding reductions are 99.6% for $A_{200}-B_n$ and 99.4% for A_n-B_{200} .

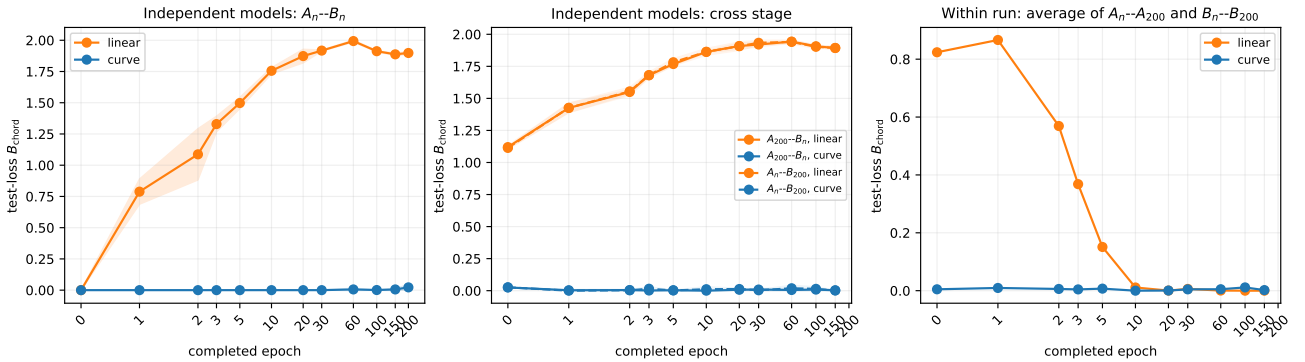


Figure 4: Test-loss B_{chord} for raw linear interpolation and optimized nonlinear paths. The panels separate same-stage independent models, cross-stage independent models, and within-run controls.

Table 2: Aggregate test-loss B_{chord} for raw linear and selected nonlinear paths.

Path group	Paths	Linear mean	Nonlinear mean
Same stage	36	1.495	0.0029
$A_{200}-B_n$	33	1.725	0.0065
A_n-B_{200}	33	1.727	0.0099
Within A	33	0.256	0.0053
Within B	33	0.253	0.0047

All 168 selected paths have sampled test-loss $B_{\text{chord}} \leq 0.05$, and 150 are below 0.02. Error is less uniformly controlled: 121 paths satisfy both loss $B_{\text{chord}} \leq 0.02$ and error $B_{\text{chord}} \leq 1.5$ percentage points. For the three final-final pairs, mean test-loss B_{chord} is 0.022 ± 0.020 , while mean test-error B_{chord} is 2.16 ± 1.95 points.

The within-run control follows a different pattern. Raw interpolation from an early checkpoint to epoch 200 has a barrier, but paths starting around epoch 10 are already close to linearly connected. Late SGD checkpoints therefore stay in a locally compatible region even though independently trained models remain incompatible without alignment or curvature.

5 Conclusions and qualifications

The current results support four conclusions:

1. Linear incompatibility between independent VGG11 runs appears within the first few epochs and later decreases; it does not grow monotonically.
2. Permutation alignment explains only part of this incompatibility. Positive ReLU scaling removes about 90% of the remaining mean test-loss B_{worse} across the checkpoint matrix.
3. B_{chord} and B_{worse} give materially different accounts of early cross-stage pairs, so both should be reported.
4. Optimized Bézier and polygonal paths avoid almost all of the raw linear test-loss barrier for the tested checkpoint pairs.

Three qualifications matter. First, the linear results use the full test set, whereas the current nonlinear profiles use a fixed 2,000-example test subset; the nonlinear selections should be re-evaluated on the full test set before a single direct comparison figure is used in the paper. Second, the original nonlinear final-final pilot gate was not met because its error threshold had been inherited from a VGG16 experiment; proceeding with the VGG11 paths was an explicitly recorded decision. Third, a found path is evidence of connectivity for the sampled data and grid, while a failed search would not prove that no path exists.

The next useful step is evaluation-only: keep all selected nonlinear paths fixed and evaluate them on the same data used by the linear report. No path refitting is required.

Artifacts

The report uses:

- `results/dense_linear_stage_vgg11_10k/;`
- `results/nonlinear_stage_vgg11_fullfit/.`

Its figures can be regenerated with `make_figures.py` in this directory.